RF-Free Quantum Magnetometry for Remote Sensing

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Introduction

- NV centers in diamond serve as spin-1quantum sensors capable of detecting magnetic fields via ODMR
- ODMR requires resonant RF fields (~2.87 GHz) to induce transitions between |ms = 0⟩ and |ms = ±1⟩
- RF driving limits scalability and integration, particularly in remote, miniaturized, or high-conductivity environments

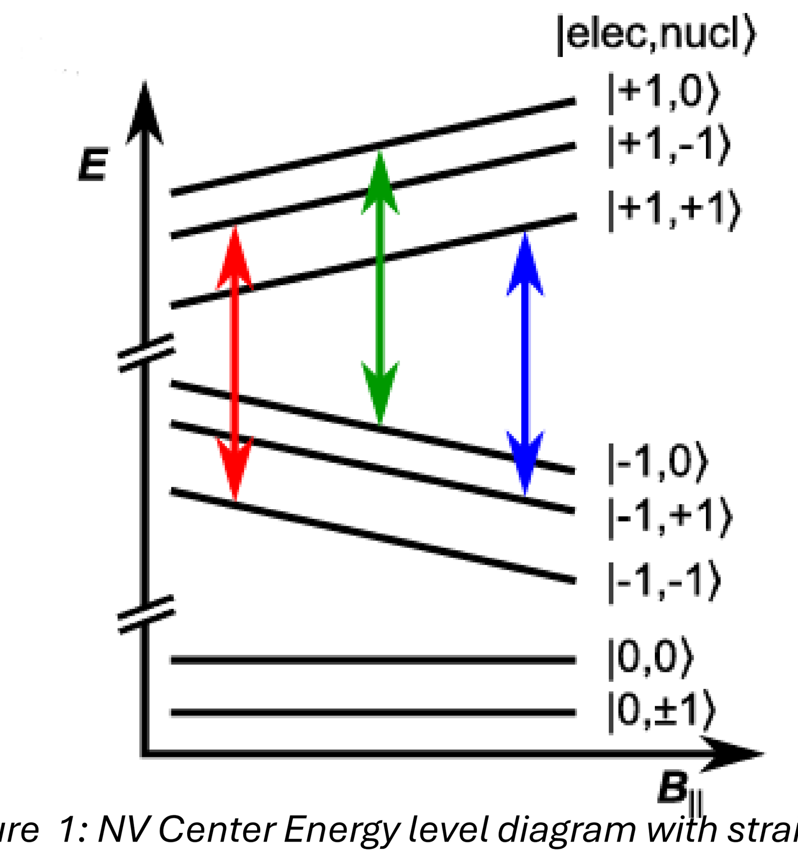


Figure 1: NV Center Energy level diagram with strain induced transitions. Sourced from mechanical spin control paper (2. SPIE (2020])

Strain

- The NV center spin Hamiltonian is sensitive to crystal strain, enabling spin manipulation via lattice deformation

$$H = DS_z^2 + g\mu_B(B_xS_x + B_yS_y + B_zS_z) + A(S_xI_x + S_yI_y + S_zI_z) + E(S_x^2 - S_y^2)$$

- Recent work suggests mechanical or thermal modulation can induce transitions without magnetic fields, known as phonon-driven magnetic resonance(PDMR)

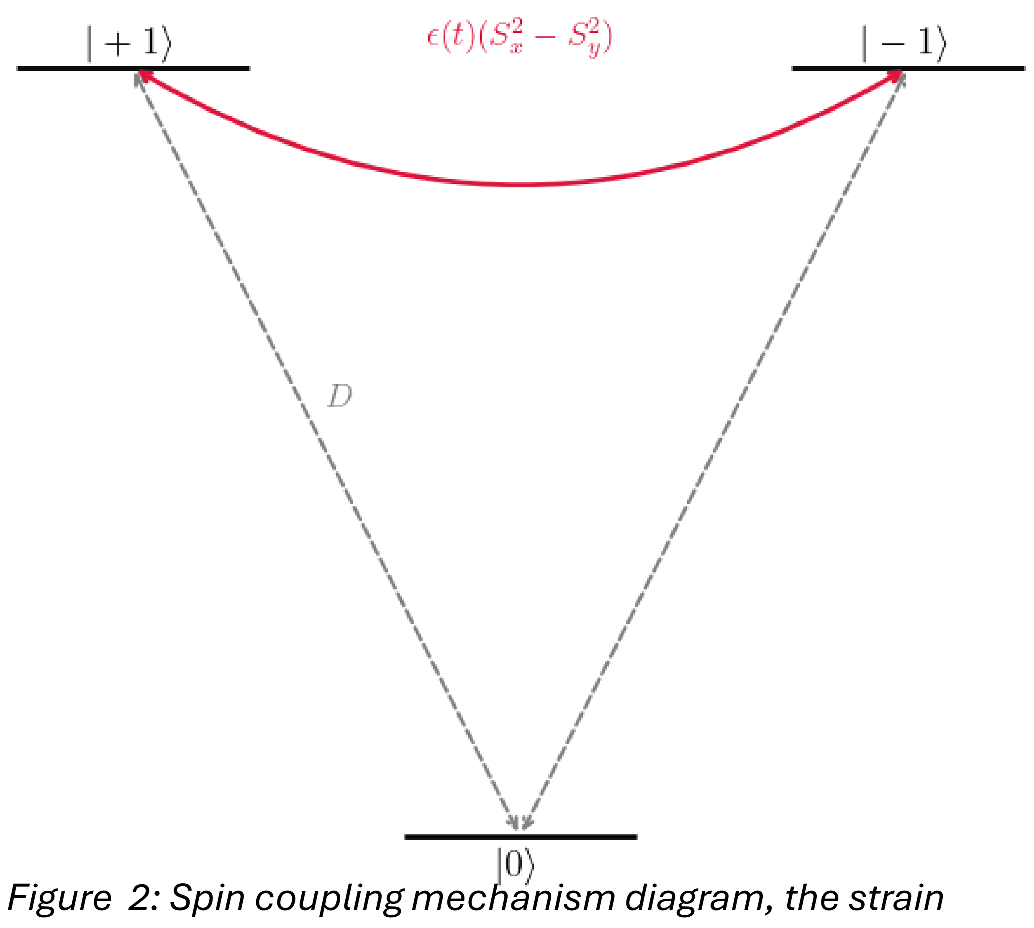


Figure 2: Spin coupling mechanism diagram, the strain enables transitions between the ± 1 states

- The strain field ϵ(t) modulates the zero-field splitting term, enabling ODMR-like spin transitions
- This approach is all-optical and fiber-compatible- removing the need for local RF circuitry or antennas

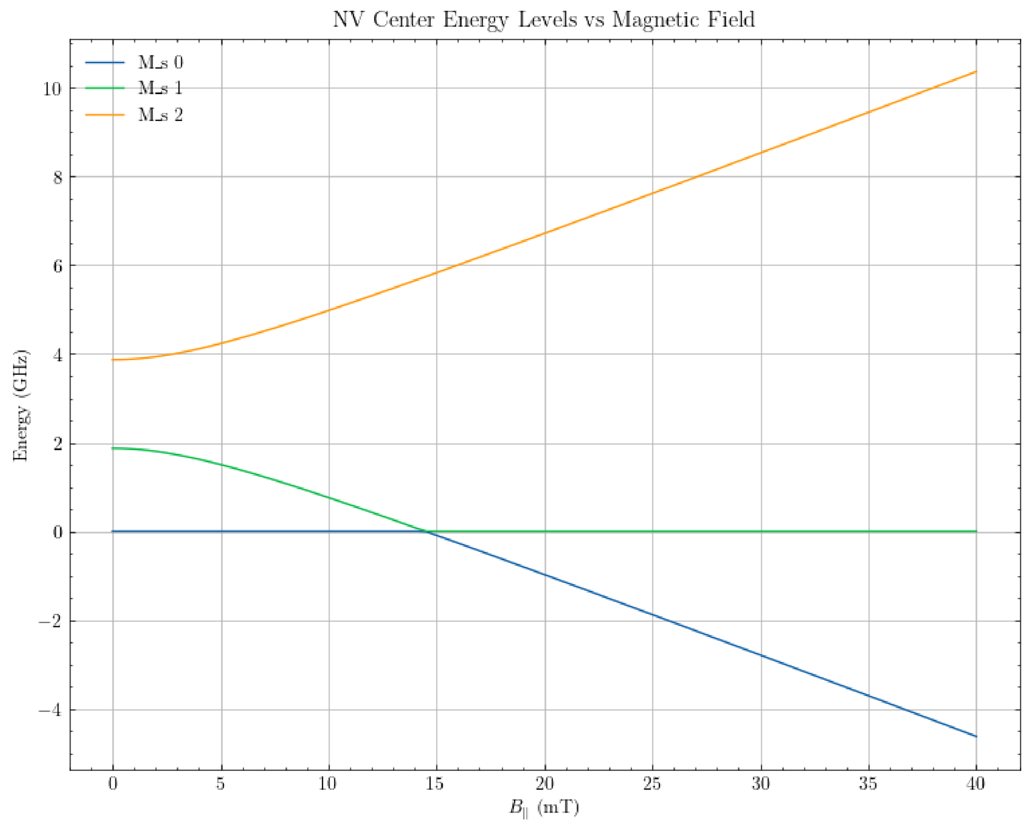
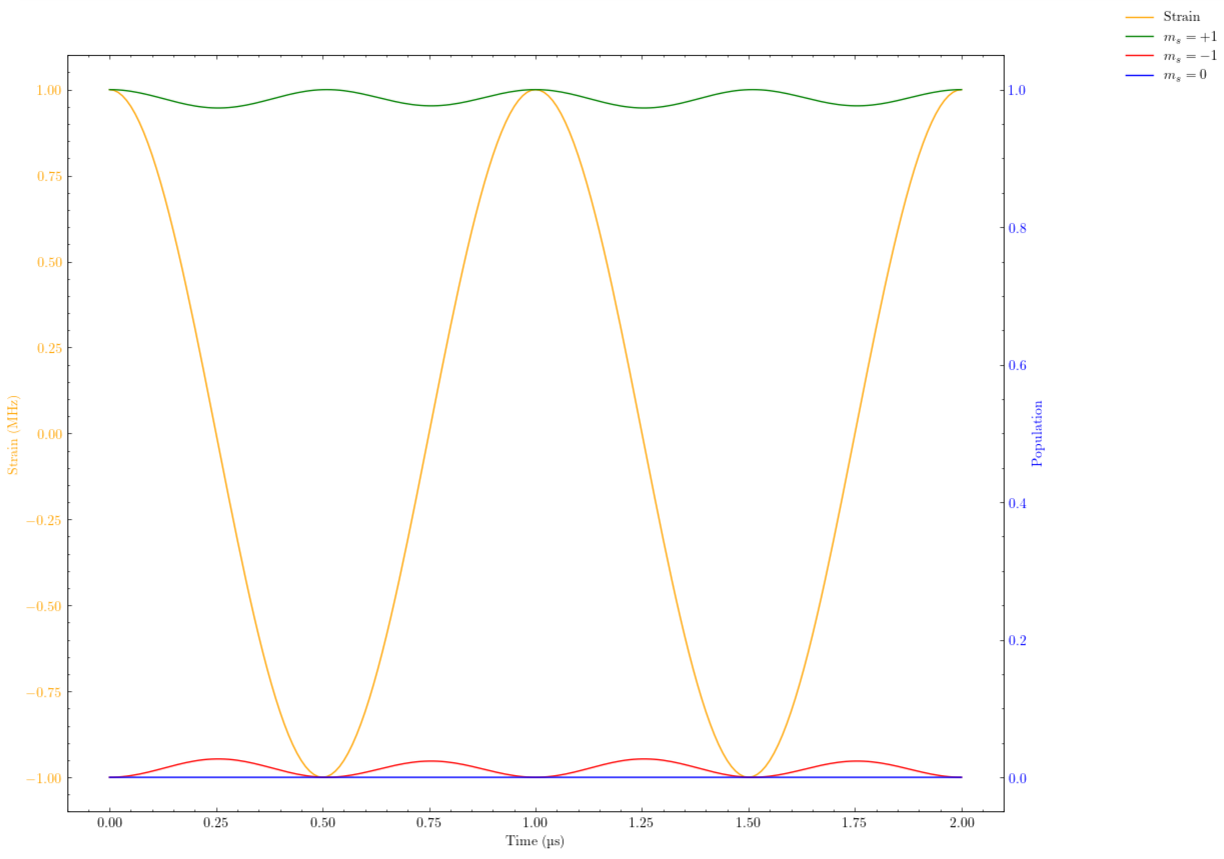


Figure 3: Computational model of

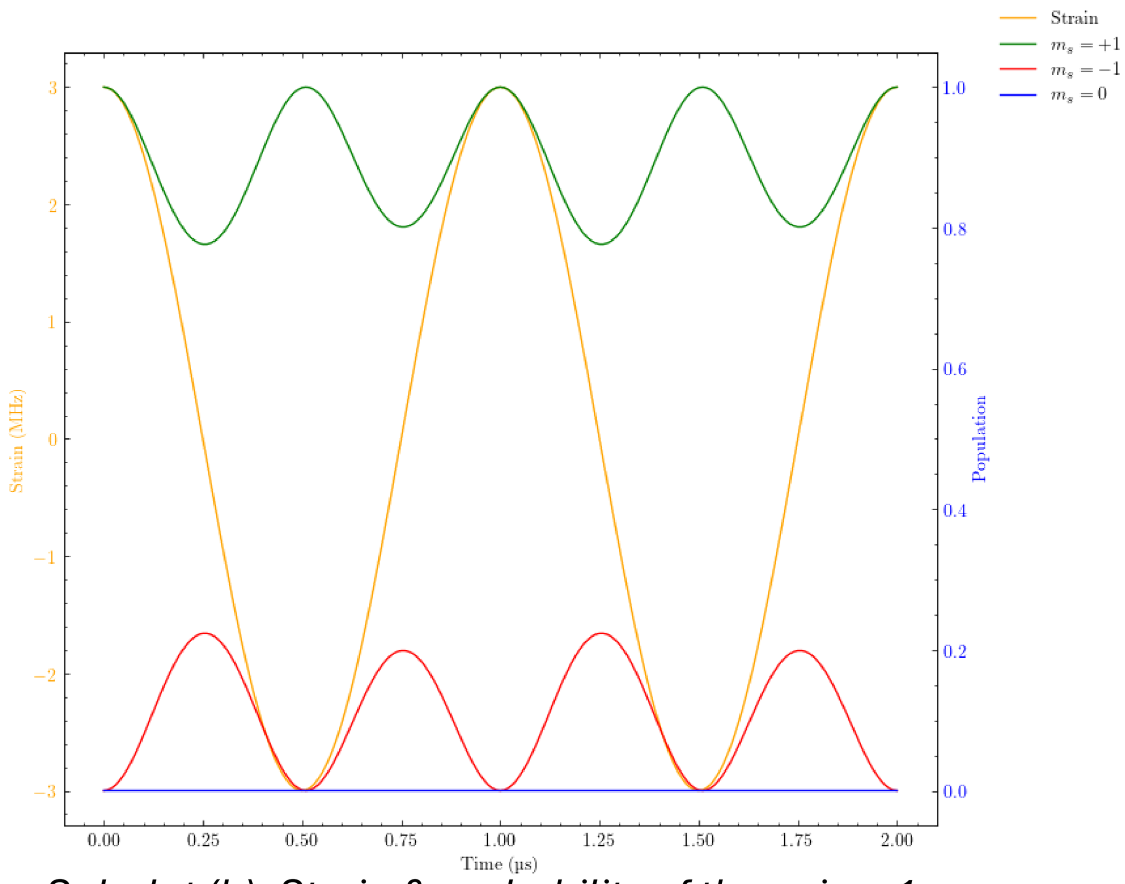
Methodology

- Computational simulation of NV center spin Hamiltonian evolution under mechanical strain control
- Systematic investigation of transition dynamics by varying strain coupling strength, phonon frequency, and magnetic field parameters
- Quantification of phonon-driven transition probabilities to assess feasibility for accessing magnetically achieved spin transitions

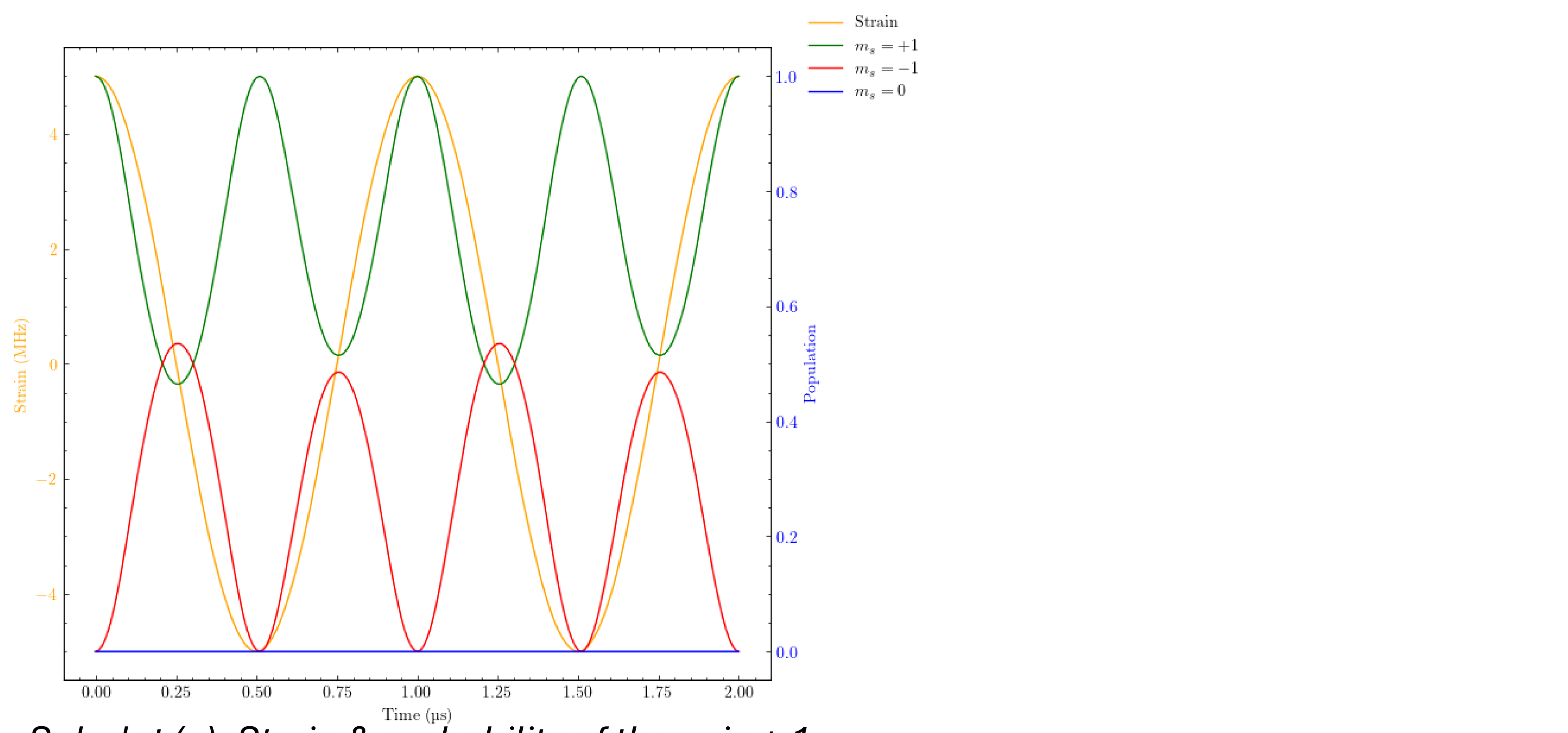
Results



Subplot (a): Strain & probability of the spin ± 1 states over time with a strain of 1Mhz



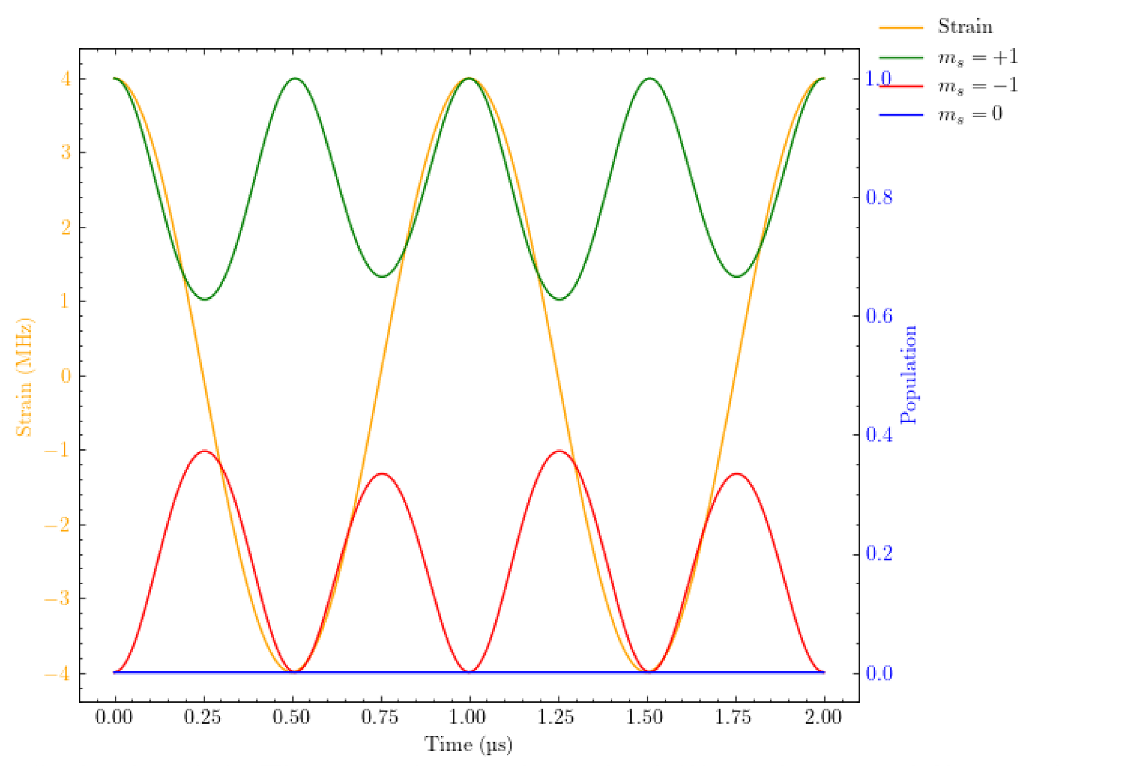
Subplot (b): Strain & probability of the spin ± 1 states over time with a strain of 3Mhz



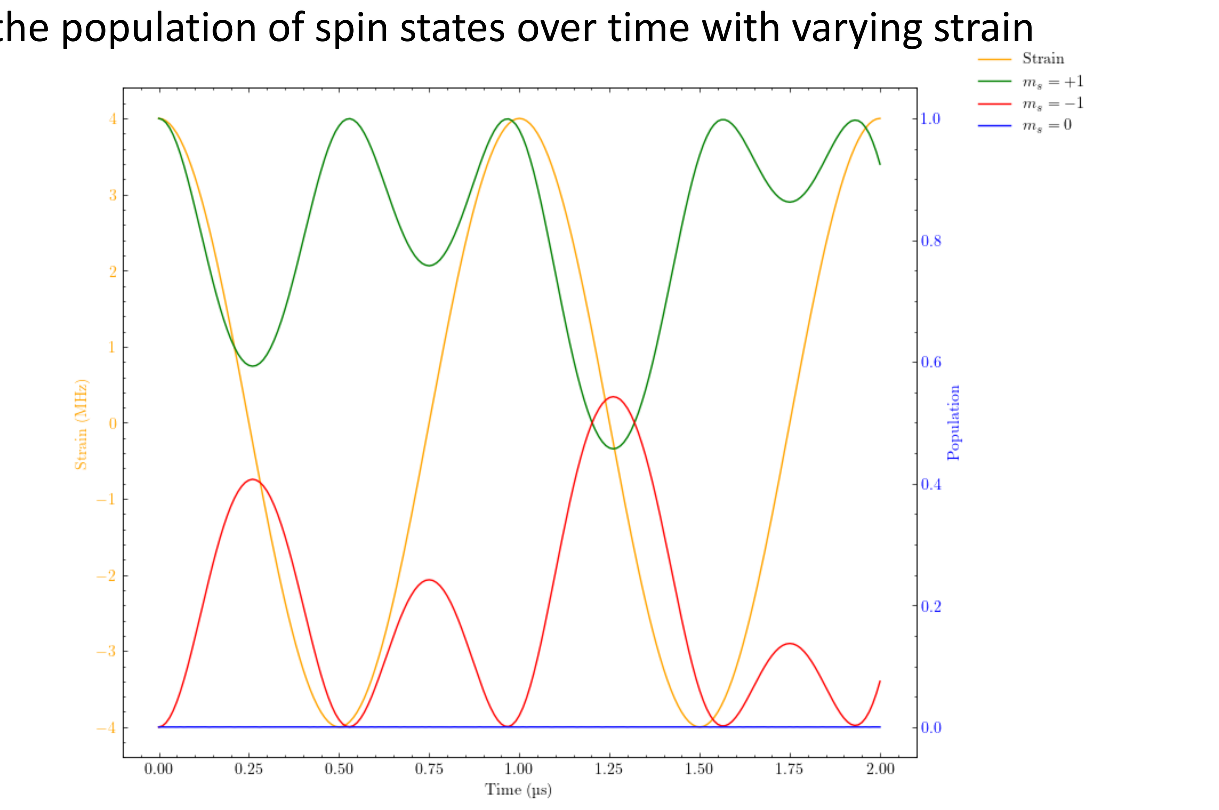
Subplot (c): Strain & probability of the spin ± 1 states over time with a strain of 6Mhz

Figure 4: Computational simulation of NV center spin Hamiltonian evolution under varying mechanical strain control

- The simulation currently can model transitions between the population of spin states over time with varying strain



Subplot (a): The strain and probability plot at 4Mhz with no magnetic field



Subplot (b): The strain and probability plot at 4Mhz with a 10 Gaussian magnetic field, forcing transition between population

Figure 5:
- Inducing a magnetic field shows forced changes in the populations of each eigenstate over time, showing that the simulation can effectively apply both strain and a magnetic field to achieve spin transitions

Conclusion

- The computational simulation of NV center spin Hamiltonian is functional however, further testing is necessary to sweep for resonant phenomena and to determine the magnitude of phonon-driven transition viability
- Further expansions to the simulation need to be made to determine what frequencies are resonant and their magnitude, the expansion from a 3 dimensional to 9 dimensional system by including the hyperfine splitting constants could further advance the simulation.

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